

4.2.2.

$$(1) F = (A+B)(A\bar{B}) = A\bar{B} + BA\bar{B} = A\bar{B}$$

$$(2) F = A + ABC + A\bar{B}C + BC + \bar{B}C = A + A(BC + \bar{B}C) + C(B + \bar{B}) = A + A + C = A + C$$

$$(3) F = \overline{\bar{A}C + BC + A\bar{B}C} + \overline{\bar{A}BC + \bar{B}C + AC} = \overline{C(\bar{A} + B + A\bar{B})} + \overline{C(\bar{A}B + \bar{B} + A)} \\ = \overline{C(\bar{A} + B)} + \overline{C(\bar{B} + A)} = \overline{CA} + \overline{CA} = \bar{C} + A + \bar{C} + \bar{A} = \bar{C}$$

$$(4) F = AD + \bar{A}\bar{D} + AB + \bar{A}C + BD + \bar{A}\bar{B}EF + \bar{B}EF$$

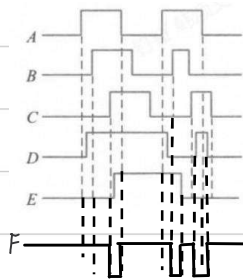
$$= A + AB + \bar{A}C + BD + \bar{B}EF = A + \bar{A}C + BD + \bar{B}EF = A + C + BD + \bar{B}EF$$

$$(5) F = \overline{\bar{A} + B} + \overline{A + \bar{B}} + \overline{\bar{A}\bar{B}\bar{A}\bar{B}} = \overline{\bar{A} + B} + \overline{A + \bar{B}} = \bar{A}\bar{B} + \bar{A}\bar{B} = \bar{A}\bar{B} + \bar{A}\bar{B} = 1$$

$$(6) F = \bar{A}D(A + \bar{D}) + ABC + CD(B + C) + A\bar{B}\bar{C}$$

$$= ABC + BCD + CD + A\bar{B}\bar{C} = AB + CD$$

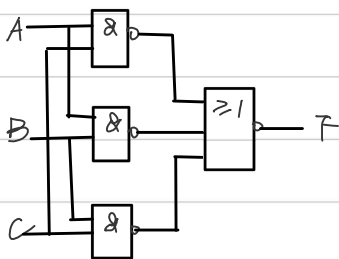
$$(7) F = (A + \bar{A}C)(A + CD + D) = (A + C)(A + D) = A + CD$$



4.3.2

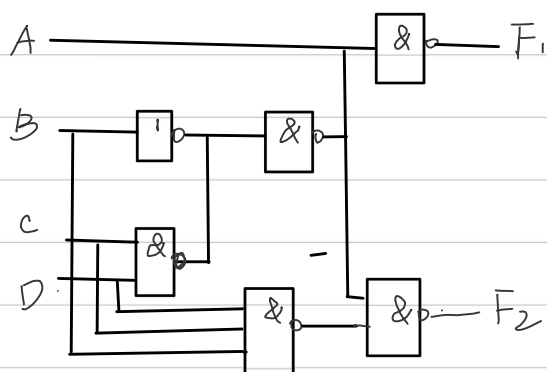
4.4.1.

$$(a) F = \overline{\bar{A}B} + \overline{\bar{B}C} + \overline{\bar{A}C}$$



A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

1b)



A	B	C	D	F ₁	F ₂
0	0	0	0	0	1
0	0	0	1	0	1
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	0	0
0	1	0	1	0	0
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	0	1
1	0	0	1	0	1
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	1	0
1	1	0	1	1	0
1	1	1	0	1	0
1	1	1	1	1	1

4.4.2.

$$1a) F = (\overline{A+B}) + (A + \overline{A+B}) + (B + \overline{A+B})$$

$$= \overline{A+B} + \overline{A}(A+B) + \overline{B}(A+B) = \overline{A+B} + \overline{A}B + \overline{B}A = (A+B)\overline{A}\overline{B}$$

$$= (A+B)(A+\overline{B})(\overline{A}+B) = (A+\overline{A}\overline{B}+AB)(\overline{A}+B)$$

$$= A(\overline{A}+B) = AB$$

$$1b) F = \overline{A+B} + \overline{B+C} + \overline{A+C} = (A+B)(B+C)(A+C) = AB+BC+AC$$

4.4.4.

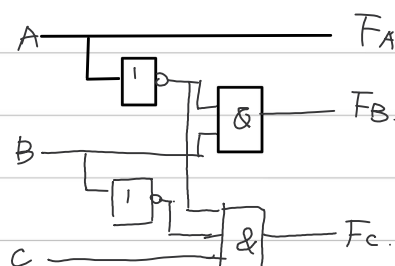
列出真值表.

A	B	C	F _A	F _B	F _C
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	0
1	0	0	1	0	0

$$\therefore F_A = A$$

$$F_B = \overline{A}B$$

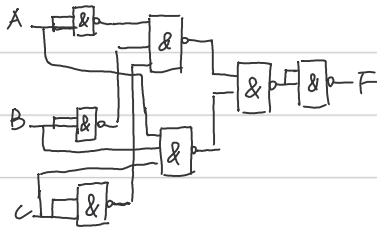
$$F_C = \overline{A}\overline{B}C$$



1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

4.4.5

$$F = \overline{ABC + \bar{A}\bar{B}\bar{C}} = \overline{\overline{ABC} \overline{\bar{A}\bar{B}\bar{C}}}$$



4.4.8

$$W = D_0 \bar{A}_1 \bar{A}_0 + D_1 \bar{A}_1 A_0 + D_2 A_1 \bar{A}_0 + D_3 A_1 A_0$$

A_1	A_0	W
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

4.4.9

$$D_0 = \bar{A} \cdot \bar{A}_0 D \quad D_1 = \bar{A} \cdot A_0 D \quad D_2 = A \cdot \bar{A}_0 D \quad D_3 = A \cdot A_0 D$$

A	A_0	D_0	D_1	D_2	D_3
0	0	D	0	0	0
0	1	0	D	0	0
1	0	0	0	D	0
1	1	0	0	0	D